1 2 3 4 5 6 7 8 9 10

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Question 1 / 10 Flag for review • 1 mark

What does the Generator in a GAN receive as input?

Options

A. Real images from the training dataset

B. Features extracted from real images

C. Output from the discriminator

D. Random noise vector from a latent space

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Question 2 / 10 Flag for review • 1 mark

Given a latent space with 8 dimensions, where each dimension can take 10 discrete values, how many unique latent vectors (and thus potentially unique generated images) are possible?

Options

A. 80

B. 18

C. 10^8

D. 1024

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Question 3 / 10 Flag for review • 1 mark

What is the ideal balance in discriminator and generator strength during GAN training?

Options

A. The discriminator wins all the time.

B. The generator wins all the time.

C. The discriminator is strong enough to challenge the generator but weak enough to be fooled sometimes

D. The discriminator and generator do not affect each other.

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Question 4 / 10 Flag for review • 1 mark

At Nash equilibrium in GAN training, which of the following statements is true?

Options

A. Generator loss = 0

B. Discriminator accuracy = 100%

C. Discriminator output ≈ 0.5 for both real and fake data

D. Generator output = random noise

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Question 5 / 10 Flag for review • 1 mark

In a GAN:

- D(x) = Discriminator output on real samples
- D(G(z)) = Discriminator output on generated samples

Consider the following possible observations after a few epochs:

Case	D(x)	D(G(z))	Likely Training Stage
(a)	0.9	0.1	?
(b)	0.6	0.4	?
(c)	0.5	0.5	?
(d)	0.9	0.9	?

Match each case (a), (b), (c), (d) respectively with the most likely training stage from below:

1.Generator producing realistic outputs; near Nash equilibrium
2.Discriminator is dominating (too strong)
3.Generator is producing outputs that fool D temporarily
4.Balanced early training where both improve steadily

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Options

A. 1342

B. 4213

C. 2431

D. 2413

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Question 6 / 10 Flag for review • 1 mark

You trained four different GANs. For each, you recorded the diversity of generated samples (measured as a normalized diversity score between 0 and 1):

Model	Diversity (0 = low, 1 = high)
A	0.91
B	0.15
C	0.95
D	0.22

Which model(s) most likely suffer from "mode collapse"?

Options

A. A and B

B. B, C, and D

C. B and D

D. All four

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Question 7 / 10 Flag for review • 1 mark

Consider the standard training loop where the discriminator trains before the generator:

```
for epoch in range(num_epochs):
    train_discriminator()
    train_generator()
```

Now, consider the following simplified GAN training loop where the generator is trained before the discriminator in each iteration:

```
for epoch in range(num_epochs):
    train_generator()
    train_discriminator()
```

What is the most likely effect of switching the training order as shown in the second loop?

Options

A. Generator tries to fool an outdated discriminator. May cause unstable training and slower convergence.

B. Changing order has no impact. Training proceeds normally with same convergence speed.

C. Discriminator becomes less accurate. Generator quickly improves and forces Nash equilibrium early.

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Question 8 / 10 Flag for review • 1 mark

A GAN has the following architecture:

Generator (G):

- Latent space dimension = 200
- One hidden layer = 1000 neurons
- Output image size = 100 × 100

Discriminator (D):

- One hidden layer = 1000 neurons
- Sigmoid activation at the output layer.

Assume fully connected layers and ignore biases.

How many parameters (in millions) will be learned by each network?

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Options

A. Generator: 5.6 M, Discriminator: 10.2 M

B. Generator: 10.0 M, Discriminator: 5.6 M

C. Generator: 20.0 M, Discriminator: 10.0 M

D. Generator: 10.2 M, Discriminator: 10.0 M

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Question 9 / 10 Flag for review • 1 mark

Common Data for Questions 9 to 10

Create a Generator class which inherits from nn.Module implementing a network which takes a latent vector of dimension 16 and pass it to a 32 dimensional hidden layer followed by ReLU and then pass it to output layer which outputs an image of size 10×10 followed by a tanh. Now sample one latent vector from torch.randn and pass it to the Generator to produce fake image.

(Use seed 42 wherever needed for reproducibility and train on CPU only. Take batch size of 1 as we are only using one image whenever needed.)

What is the sum of all the elements of output image by Generator? Enter your answer upto three decimal places.

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Your answer

Enter your answer

Only numeric input is accepted for this question.

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Question 10 / 10 Flag for review • 1 mark

Now create a Discriminator class with the below setup:

• Linear(100 -> 32)

• ReLU()

• Linear(32 -> 1)

The input to this is the flatten image of 1×10×10 (from your previous Generator). The output is without sigmoid at the end (logits output). What is the output of this Discriminator? Enter your answer upto three decimal places.

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Your answer

Enter your answer

Only numeric input is accepted for this question.

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